



TOWARDS ROBUST MULTIMODAL EMOTION RECOGNITION IN CONVERSATIONS: AUDIO, TEXT, AND UNCERTAINTY ESTIMATION

Bachelor's Project Thesis

Teodora Stereciu, s4678826, t.stereciu@student.rug.nl,

Supervisor: Dr Tsegaye Misikir Tashu

Abstract:

Emotion recognition has a wide range of applications across domains, including market research, health care, and education. Text-based approaches are limited because natural language is dynamic and nuanced, especially in informal conversations. Emotion recognition systems leveraging visual, acoustic, or physiognomical cues are often more effective and robust. This study explored the effect on performance and uncertainty estimation of using a multimodal model with text and audio data over single modality classifiers for emotions in conversational data. It compared early, intermediate, and late fusion strategies of BERT text encodings and emobase functional features of the audio. The best-performing model was text-only, with late fusion closely following. Multimodality showed positive effects on the confidence estimates. The results also suggest difficulties in processing emotions from audio, which may be attributed to high complexity and data noise. Future research may improve on the contributions of this study by addressing the over-representation of neutral expressions and modelling the conversational context.

1 Introduction

With the rise of computer-mediated communication, many emotion recognition applications surfaced. Human emotion is used across several domains, including in market research, health care, and education (Dzedzickis et al., 2020). Text-based approaches to emotion recognition are limited because natural language is dynamic and nuanced, especially in informal conversations. For example, the sentence “That’s just great.” could be interpreted as genuine enthusiasm or sarcastic frustration, depending on the context, tone, and accompanying facial expressions. A text-based model may need help discerning these nuances and misinterpret the intended emotion.

A single-modality system, in general, is less robust to noise and data quality issues. Alternatively, models can leverage more than one input modality. Combining information from multiple modalities to improve the analysis of a particular problem is the process of multimodal fusion. It involves integrating data from different sources, such as text, im-

ages, audio, video, and sensor readings, to attain a more precise and complete representation of the underlying information (Pawlowski et al., 2023).

Multimodal fusion is instrumental in emotion recognition since various indicators of a textual, visual, acoustic, or physiognomical type may be available. Needless to say, it is also essential to ensure reliability in real-life emotion recognition applications, such as in telemedicine and remote mental health therapy sessions. Automated emotion recognition may be used in this context to assist the work of health practitioners. If the audio quality is poor due to technical issues, the patient’s statements may still provide substantial information for analysis. In contrast, if the text is ambiguous or lacks emotional depth, the audio component can reveal nuances through vocal cues such as tone, pitch, and pace of speech.

Although multimodality improves the security of automated models, it does not make them fool-proof. Emotion recognition is highly subjective because it depends on sociocultural factors. It makes

assumptions about how emotions are expressed, which may be invalid in specific contexts (Waelen, 2024). As a result, it is virtually impossible to train a model that can generalize across any scenario since its knowledge comes from data where ground truth is not absolute but decided by human judges. A data-driven strategy is also not guaranteed to be correct when extrapolating to an unknown domain. To understand and mitigate the effect of these data and model limitations, a system can be adapted to quantify the uncertainty of each prediction (Abdar et al., 2021). If the model detects an ambiguous case (i.e., with high uncertainty), it can refrain from offering a (misleading) result and redirect it for inspection to a human expert. In this way, uncertainty estimation makes the process safer and more transparent.

Taking into account the benefits of multimodal fusion and uncertainty estimation discussed above, this research aims to investigate the application of both techniques to emotion recognition in conversations where text and audio are available. Specifically, this study seeks to answer the following question.

How does fusing text and audio features from conversational data affect emotion recognition performance and uncertainty estimates compared to text-only and audio-only approaches?

The following sections will begin by presenting related work in emotion recognition from text and audio and in predictive uncertainty quantification. Then, the specific approach to multimodal fusion and uncertainty estimation will be described, along with implementation and experimental details. The paper will report the results and discuss them in the overall context of the current project and general task, and lastly close with recommendations for future research.

2 Background

2.1 Related work in emotion recognition

Previous research into unimodal emotion recognition focused on four broad channels: visual, vocal, textual, and physiological. In their review, Singh

Tomar et al. (2023) found that visual models, primarily based on facial expressions, tend to be the most effective and practical in real use cases. However, they suffer from noisy sensors, image processing issues due to high dimensionality, and privacy concerns.

Similar performance to visual cues can be obtained with the vocal modality, which covers speech, prosody (i.e., information that comes from sound duration, pitch, and intensity), and intonation methods. Recently, audio classifiers based on hand-crafted features (Eyben et al., 2013; Schuller et al., 2013) received less attention, and more focus has been given to end-to-end models directly matching audio spectrogram representations to their labels. Convolutional neural networks (CNNs; O’Shea & Nash, 2015), specifically, have been leveraged for their spatial locality and translation equivariance qualities (Trigeorgis et al., 2016). A similar, hybrid approach better captures long-range global context with self-attention mechanisms on top of the CNN (P. Li et al., 2018). Gong et al. (2021) also proposed a convolution-free, purely attention-based model, which outperformed CNNs on specific tasks. These advancements are subject to a trade-off between performance and interpretability (Dziugaite et al., 2020). As a more transparent alternative, feature sets developed for emotion recognition (Doğdu et al., 2022) can still achieve competitive performance in a given domain without the added complexity of end-to-end models (Geethu et al., 2023).

Noise and complexity issues may also affect audio signals, whereas textual inputs are more straightforward and more economical by comparison (Singh Tomar et al., 2023). Recurrent neural networks (RNNs) are prevalent in text emotion recognition (Chatterjee et al., 2019; TGDK et al., 2021; Y. Li et al., 2023), thanks to their ability to cover long-term dependencies. Recently, transfer learning has been widely adopted to draw on the predictive abilities of transformer large language models (Gandhi et al., 2022).

2.2 Multimodal fusion

It has been proposed that more than one modality for emotion recognition be used to overcome the limitations of each modality. Audio-text fusion is a popular strategy that has enhanced emotion de-

tection from speech (Yoon et al., 2018; Pepino et al., 2020; Y. Li et al., 2022). Overall, there are four primary fusion strategies: early, intermediate, late, and hybrid fusion. They differ in how much processing is done on the modalities before and after fusion (e.g., an early fusion model learns on the joint representation of the features, while late fusion processes them independently until the final prediction). Depending on data characteristics, common fusion operations are concatenation, addition, and multiplication.

2.3 Uncertainty estimation

Analyses such as that of Abdar et al. (2021) revealed that uncertainty in machine learning arises from two fundamental sources: the data and the model. Data uncertainty, also referred to as aleatoric uncertainty, measures the natural variation and noise in all data. Aleatoric uncertainty can be either homoscedastic if it stems from the constant noise across data points or heteroscedastic otherwise. Model uncertainty, also called epistemic uncertainty, indicates a lack of knowledge about the data on the model’s part. Bases of epistemic uncertainty include model misspecification, class imbalance, and lack of training data (Abdar et al., 2021).

A popular approach to uncertainty quantification is based on Bayesian neural networks (Gawlikowski et al., 2022). Bayesian methods infer the probability distribution over the network parameters $\theta = (w_1, \dots, w_k)$. More specifically, given a training input-target pair (x, y) , the posterior distribution over the space of parameters $p(\theta | x, y)$ is modelled by assuming a prior distribution over the parameters $p(\theta)$ and applying Bayes theorem (Equation 2.1).

$$p(\theta | x, y) = \frac{p(y, x | \theta) p(\theta)}{p(y | x)} \quad (2.1)$$

Given this estimation of the posterior distribution over the weights (Equation 2.1), the prediction of output y^* for a new input x^* can be obtained by marginalizing the likelihood $p(y | x, \theta)$ with the posterior distribution, as shown in Equation 2.2.

$$p(y^* | x^*, x, y) = \int_{\theta} \overbrace{p(y^* | x^*, \theta)}^{\text{forward pass}} \underbrace{p(\theta | x, y)}_{\text{weights posterior}} d\theta \quad (2.2)$$

Equation 2.2 can be interpreted as integrating the product of a forward pass (given a specific set of weights) and the probability of that set of weights, overall weight values. Since the integral of Equation 2.2 is usually intractable for the common prior-posterior pairs, Monte Carlo approximation with M samples can be used to obtain a solution. The prediction will be given by the mean of M deterministic networks, $f_{\theta_1}, \dots, f_{\theta_M}$, parameterized by M samples, $\theta_1, \theta_2, \dots, \theta_M$, from the posterior distribution of the weights (Equation 2.3).

$$y^* \approx M^{-1} \sum_{i=1}^M y_i^* = M^{-1} \sum_{i=1}^M f_{\theta_i}(x^*) \quad (2.3)$$

3 Approach

3.1 Fusion strategies

Three fusion strategies were explored in this study: early, intermediate, and late. Figure 3.1 show how these strategies were implemented. Each strategy expects parallel text features and audio features. The fusion operation was concatenation in all strategies. In early fusion, the feature vectors are fused at the start before being further processed in a dense network with dropout and classified with uncertainty via ‘sampling softmax.’ Further elaboration on the classification with uncertainty method will be provided later in this section. The intermediate fusion model first learns lower-dimensional embeddings for each modality and then concatenates them for further processing. In the case of late fusion, each modality is mostly processed independently of the other. The processed outputs are concatenated and classification is carried out on this joint representation.

The fusion models were compared against two unimodal baselines: text-only and audio-only. Each baseline consisted of a dense network with dropout and classification with uncertainty on the respective features.

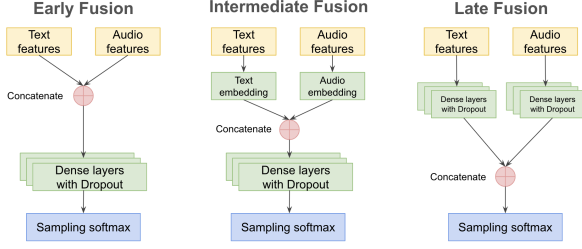


Figure 3.1: Fusion strategies explored in this study. The fusion operation was concatenation. The strategies differ in how much processing is done individually versus together in terms of the modalities.

3.2 Uncertainty estimation and disentanglement

In each of the presented strategies and baselines, aleatoric uncertainty was estimated with ‘sampling softmax’ (Kendall & Gal, 2017). This project also quantified epistemic uncertainty using Monte Carlo (MC-) Dropout (Gal & Ghahramani, 2016).

Sampling softmax is a method for classification with heteroscedastic aleatoric uncertainty, proposed by Kendall & Gal (2017). In their study of Bayesian deep learning models for vision tasks, Kendall & Gal developed a custom softmax activation layer, in which logits have uncertainty (i.e., Gaussian mean and variance, as shown in Equation 3.1). N samples are used to pass the Gaussian logit distribution $\hat{\mathbf{z}}$ through the softmax activation and produce $p(y | \mathbf{x})$ (Equation 3.2), a predictive distribution that accounts for (heteroscedastic) aleatoric uncertainty. Following Valdenegro-Toro & Mori (2022), in this paper the formulation is referred to as sampling softmax.

$$\hat{\mathbf{z}}_j \sim \mathcal{N}(\mu(\mathbf{x}), \sigma^2(\mathbf{x})) \quad (3.1)$$

$$p(y | \mathbf{x}) = N^{-1} \sum_i \text{softmax}(\hat{z}_i) \quad i \in [1, N] \quad (3.2)$$

MC-Dropout is a technique that interprets dropout training as a Bayesian approximation to estimate uncertainty in neural networks. Dropout (Srivastava et al., 2014) is a regularization method

that randomly drops neurons during training, effectively creating an ensemble of subnetworks. Gal & Ghahramani (2016) extended this concept to inference with MC-Dropout. They showed that by keeping dropout active during testing and performing multiple forward passes with different dropout masks, one can approximate the posterior distribution over the model parameters (Equation 2.3). This results in a predictive distribution that accounts for model uncertainty.

MC-Dropout was used on top of each classifier with sampling softmax to obtain predictions that also factor in epistemic uncertainty. The resulting total uncertainty was disentangled into the two components following the approach of Valdenegro-Toro & Mori (2022). For the j th forward pass with MC-Dropout, $j \in [1, M]$, the aleatoric uncertainty σ_{ale}^2 and the epistemic uncertainty σ_{epi}^2 are given by Equation 3.3, where σ_j^2 and μ_j are the mean and variance of the logits. The aleatoric variance σ_{ale}^2 combines all variances into a single variance, while the epistemic variance σ_{epi}^2 determines the disagreement between the M samples (i.e., forward passes).

$$\sigma_{\text{ale}}^2(\mathbf{x}) = \mathbb{E}_j[\sigma_j^2(\mathbf{x})], \quad \sigma_{\text{epi}}^2(\mathbf{x}) = \text{Var}_j[\mu_j(\mathbf{x})] \quad (3.3)$$

To obtain aleatoric and epistemic probabilities, the logit distributions given by the mean $\mu(\mathbf{x}) = \frac{1}{M} \sum_i \mu_i(\mathbf{x})$ and either $\sigma_{\text{ale}}^2(\mathbf{x})$ or $\sigma_{\text{epi}}^2(\mathbf{x})$ were passed through sampling softmax (Equations 3.4 and 3.5). Equation 3.6 computes the predictive probabilities with both uncertainty types.

$$p_{\text{ale}}(y | \mathbf{x}) = \text{SamplingSoftmax}(\mu(\mathbf{x}), \sigma_{\text{ale}}^2(\mathbf{x})) \quad (3.4)$$

$$p_{\text{epi}}(y | \mathbf{x}) = \text{SamplingSoftmax}(\mu(\mathbf{x}), \sigma_{\text{epi}}^2(\mathbf{x})) \quad (3.5)$$

$$p_{\text{pred}}(y | \mathbf{x}) = \text{SamplingSoftmax}(\mu(\mathbf{x}), \sigma_{\text{pred}}^2(\mathbf{x})), \quad \sigma_{\text{pred}}^2(\mathbf{x}) = \sigma_{\text{ale}}^2(\mathbf{x}) + \sigma_{\text{epi}}^2(\mathbf{x}) \quad (3.6)$$

Table 4.1: Example conversation in the dataset with text, audio, and emotion labels.

Speaker	Text	Audio	Emotion
Chandler	I never sucked , I actually did not want you to know how good I was !	dia1374utt1.wav	neutral
Monica	Why ?	dia1374utt2.wav	surprise
Chandler	I do not know .	dia1374utt3.wav	neutral
Monica	This is so great ! Now we can enter into doubles tournaments !	dia1374utt4.wav	joy
Chandler	That is why !	dia1374utt5.wav	neutral

4 Methods

4.1 Data

The dataset used in this project was the Emotion-Cause-in-Friends dataset released by Wang et al. (2021). It contains 1,374 conversations, which add up to 13,619 utterances. The authors split the conversations into train, validation, and test sets. The data were sourced from the 1994-2004 sitcom “Friends” and contain casual and familiar language between several speakers (Table 4.1).

Along with the text captions, Wang et al. provided audio and video data for each utterance in MP4 files. For this study, the audio signal was extracted from each MP4 video as a WAV file using the Moviepy library (*User Guide - MoviePy 1.0.2 documentation*, n.d.). Each utterance was annotated by human judges with the speaker’s name and one of seven emotions: Ekman’s six basic emotions (i.e., anger, disgust, fear, happiness, sadness, and surprise; Ekman & Friesen, 1971) and a seventh option, ‘neutral’ (Figure 4.1). Table 4.1 shows an example conversation between two speakers, where each conversation line has an associated text, audio, and emotion.

4.2 Data Preprocessing

4.2.1 Audio

An openSMILE feature set created for emotion recognition, ‘emobase’ (Eyben et al., 2010), was extracted from each audio signal. The emobase configuration contains 988 features, including the following low-level descriptors (LLDs): intensity, loudness, twelve Mel-Frequency Cepstral Coefficients, pitch (F_0), probability of voicing, envelope, eight

Line Spectral Frequencies, and Zero-Crossing Rate. To complete the feature set, delta regression coefficients are computed from the LLDs, and the following functionals are applied to the LLDs and their delta coefficients: maximum and minimum value and respective relative position within the input, range, arithmetic mean, two linear regression coefficients and linear and quadratic error, standard deviation, skewness, kurtosis, quartile 1–3, and three inter-quartile ranges.

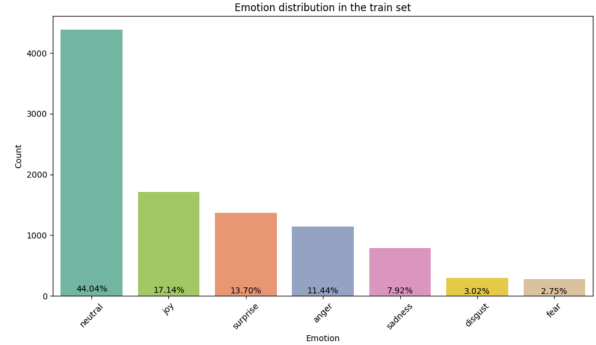


Figure 4.1: Class distribution in the train set. Almost half of the samples are labelled as ‘neutral.’ The ‘disgust’ and ‘fear’ emotions are in considerable minority in the data.

4.2.2 Text

Text features were obtained with the BERT base model, pre-trained on uncased English sentences (Devlin et al., 2018). The raw text was preprocessed as the BERT model demands, with a sequence length of 35, as suggested by Wang et al. (2021). The BERT output for the preprocessed tokens was contextualized (at the utterance level)

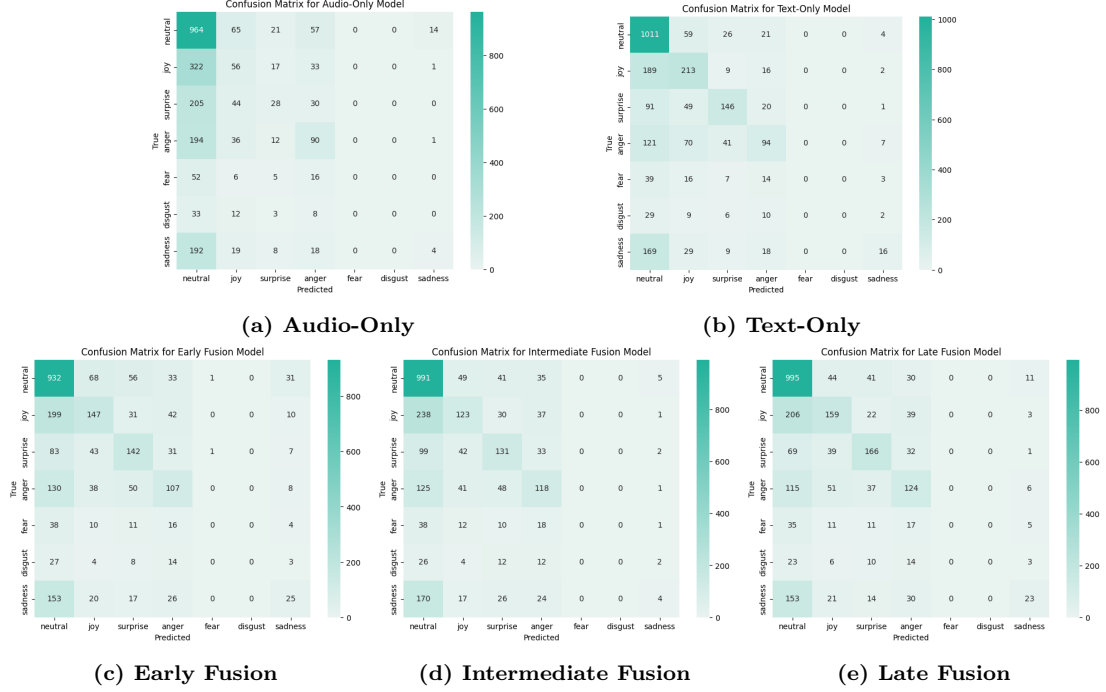


Figure 5.1: Predicted labels on the test set compared to the ground truth with unimodal and fusion approaches. No model learned to predict ‘fear’ or ‘disgust.’ The majority class, ‘neutral,’ is often predicted across all models, sometimes incorrectly.

text encodings. The encodings were extracted from the last layer’s hidden state of the first sequence token after processing through a linear layer and a *tanh* activation function.

4.3 Train settings

Every model was trained to minimize the categorical cross-entropy loss via the Adam optimizer with batches of 64 samples. Training consisted of at most 300 epochs but was interrupted after five epochs with no improvement to the validation loss by an early-stopping strategy.

4.4 Hyperparameter search

For each model (i.e., Text-Only, Audio-Only, Early Fusion, Intermediate Fusion, and Late Fusion), a random search was run on the hyperparameters to maximize the accuracy on the validation set. Specifically, the search tuned the learning rate, number of units in the dense layers, and dropout rate. Each search consisted of 10 iterations, with the train

settings explained previously. The detailed architecture of each model defined after tuning can be found in Appendix A.

4.5 Evaluation

The model predictions with total uncertainty on the test set were generated for every setup. The evaluation of each model was based on the predictions’ accuracy score and categorical cross-entropy. Classwise performance was also assessed with confusion matrices. Additionally, the predictions with disentangled epistemic ($M = 100$) and aleatoric ($N = 100$) uncertainty and their entropy were plotted for an example conversation (Table 4.1) from the test set.

5 Results

Table 5.1 shows the test results of each approach. Text-only achieved the best performance with 57.72% accuracy. Late fusion reached similar per-

Model	Accuracy	Categorical Cross-Entropy
Audio-Only	44.19%	1.53
Text-Only	57.72%	1.25
Early fusion	52.73%	1.34
Interm. fusion	53.51%	1.30
Late fusion	56.98%	1.25

Table 5.1: Model performance on the test set. The best-performing model was text-only, closely followed by late fusion.

formance. Audio-Only had the worst performance of all approaches at 44.19%.

Confusion matrices (predicted versus true labels) for the test predictions of every model (Figure 5.1) offer a more detailed look at class-specific performance. The minority classes (‘fear’ and ‘disgust’) do not appear to have been learned successfully in any approach, since no model actually predicted them for the test set. Across models, there is also a bias in predicting the majority class, ‘neutral.’

Additionally, to gain insights into the uncertainty

estimates, the disentangled predictions for an example conversation (Table 4.1) are presented in Figures 5.2, 5.3, 5.4, 5.5, and 5.6.

6 Discussion

The experiments of this project concerned five models: two unimodal (audio-only and text-only) and three with multimodal fusion (early fusion, intermediate fusion, and late fusion). Each model was trained on conversational data labelled with one of seven emotions. The classification of every setup accounted for aleatoric uncertainty via a custom softmax layer, i.e., sampling softmax, wherein class logits have uncertainty. MC-Dropout was used to estimate epistemic uncertainty. At inference time, the total uncertainty was disentangled for analysis.

The evaluation of the unimodal approaches (text-only and audio-only) as well as different fusion strategies (early, intermediate, and late) revealed precise emotion recognition to be challenging. Audio data, in particular, had limited utility for the task. The audio-only model performed poorly and



Figure 5.2: Predicted probabilities with disentangled uncertainty for a neutral conversation line, “I never sucked; I actually did not want you to know how good I was !” For each unimodal or multimodal approach, the probabilities with predictive, aleatoric, and epistemic uncertainty are presented. The entropy of each of these distributions is displayed in the upper right corner of each subplot. No model makes the correct prediction, misclassifying it as ‘anger’ with varying confidence levels. Early fusion, in particular, showed the highest degree of overconfidence (predictive entropy 1.25). On the other hand, audio-only made the most uncertain prediction (predictive entropy 1.80).

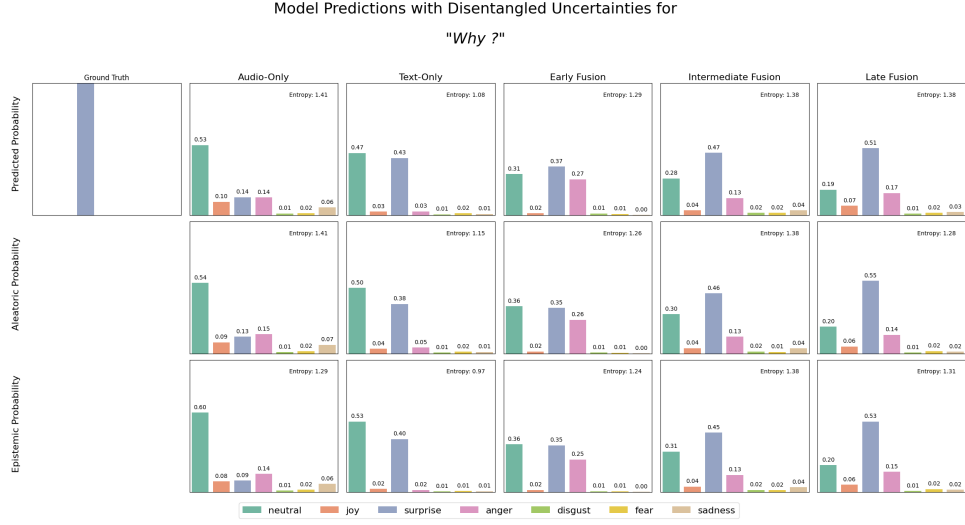


Figure 5.3: Predicted probabilities with disentangled uncertainty for a surprised conversation line, “Why ?” The probabilities with predictive, aleatoric, and epistemic uncertainty are presented for each unimodal or fusion approach. The entropy of each of these distributions is displayed in the upper right corner of each subplot. Unimodal approaches misclassify the example as ‘neutral.’ Fusion models can make the correct prediction, with late fusion (rightly) being the least uncertain.

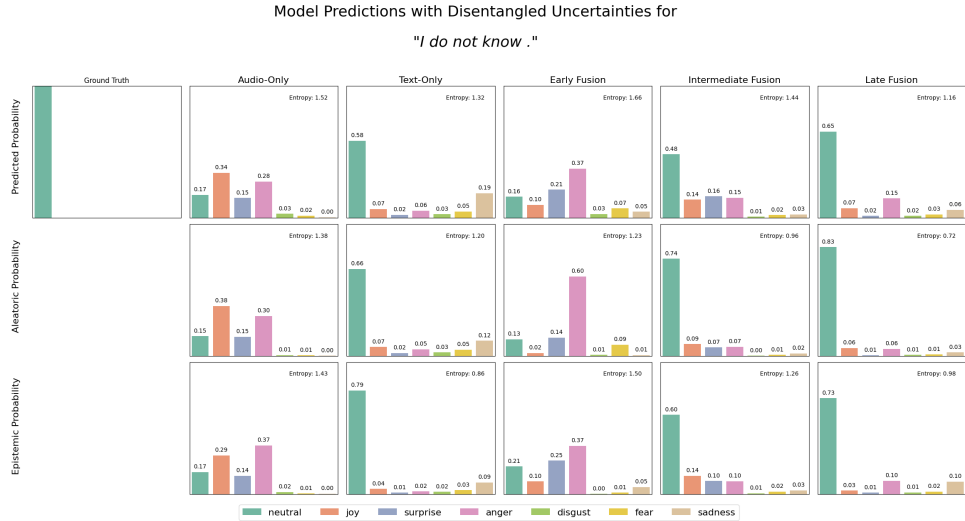


Figure 5.4: Predicted probabilities with disentangled uncertainty for a neutral conversation line, “I do not know .” The probabilities with predictive, aleatoric, and epistemic uncertainty are presented for each unimodal or fusion approach. The entropy of each of these distributions is displayed in the upper right corner of each subplot. Audio-only and early fusion make incorrect predictions (i.e., ‘joy’ and ‘anger’).

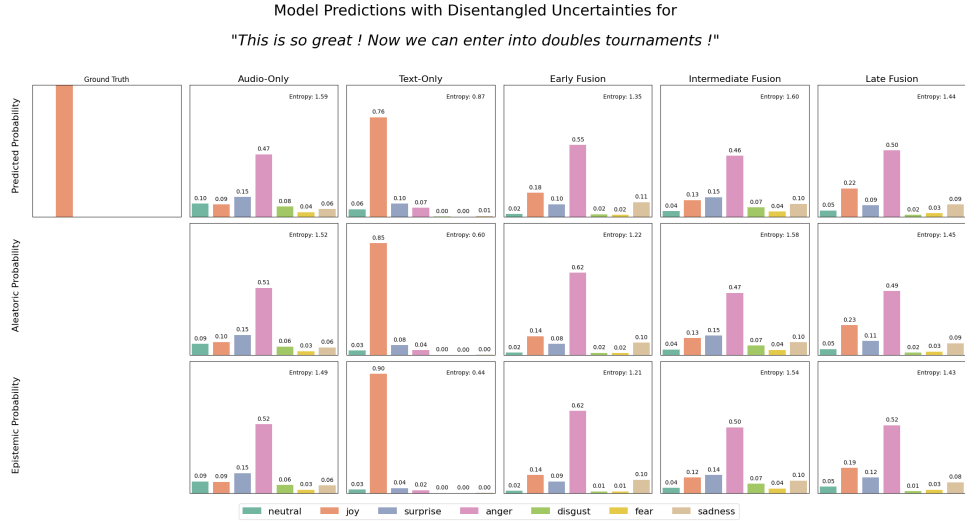


Figure 5.5: Predicted probabilities with disentangled uncertainty for a joyful conversation line, “This is so great! Now we can enter into doubles tournaments !” The probabilities with predictive, aleatoric, and epistemic uncertainty are presented for each unimodal or fusion approach. The entropy of each of these distributions is displayed in the upper right corner of each subplot. Text-only is the single model used to classify this example correctly. All other approaches predict ‘anger’.

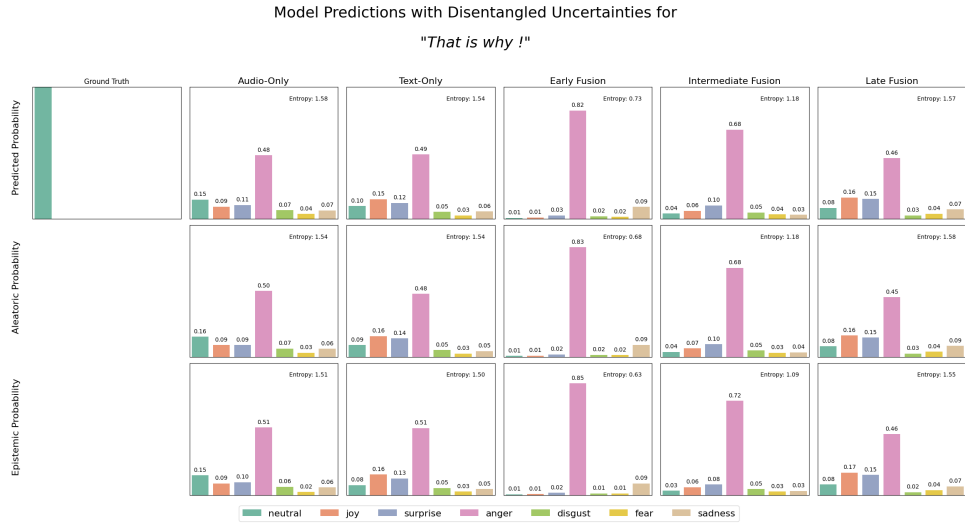


Figure 5.6: Predicted probabilities with disentangled uncertainty for a neutral conversation line, “That is why !” The probabilities with predictive, aleatoric, and epistemic uncertainty are presented for each unimodal or fusion approach. The entropy of each of these distributions is displayed in the upper right corner of each subplot. All models misclassify the example as ‘anger.’ Early fusion is especially ineffective because it has very high confidence.

mostly learned to exploit the class imbalance and predict the majority class (Figure 5.1a). In fact, the aleatoric uncertainty analysis revealed potential noise or complexity in the audio features (Figures 5.2, 5.4), which indicates the audio signal may have been unreliable (i.e., too noisy) and the emobase feature set was not well-suited for this raw data.

All fusion strategies outperformed audio-only, but ultimately reached modest results. Overall, the class imbalance in the data heavily impacted the performance of every model. The evaluation outcomes suggest that more independent processing on the two modalities is beneficial, although this finding may have been due to the fact that the fusion models mostly rely on the text features and fail to effectively mine the audio. The unimodal text classifier was marginally better than the best performing fusion system, but was still limited by ambiguities and lack of conversational context (Figures 5.2, 5.3). Some insights from the audio features could disambiguate such an example (Figures 5.3), demonstrating the benefit of multimodality.

7 Conclusion

This study applied multimodal fusion and uncertainty estimation to an emotion recognition in conversations task. With the spread of applications of human emotion detection, it becomes necessary to ensure system reliability against model and data limitations. A possible way to make emotion recognition more robust is by leveraging multiple input modalities, allowing cross-referencing and disambiguation. Additionally, determining each prediction’s confidence may help manage risks and expectations regarding the system. The findings of this paper suggest that indeed multimodality can improve and make emotion recognition more confident, despite data limitations in the experiment. It should also be noted that certain modalities carry more significant insights for emotion detection from conversations, inherently but also due to recurring problems with noise.

8 Future work

Future research may address the class imbalance issue, which is present in several popular datasets for multimodal emotion recognition (Busso et al., 2008; Poria et al., 2019), by applying data augmentation methods that take into account multiple modalities (e.g., Meng et al., 2023). Another critical limitation of this study was that it did not model the conversational context, including speaker information, which would clarify the intended emotion behind each utterance. Perhaps similar approaches for dialogue systems as that of Majumder et al. (2019), Ghosal et al. (2019), and Hu et al. (2021) would be beneficial. Finally, uncertainty estimation in machine learning models is still an underexplored area of research. The advantages of specific methods over others have yet to be fully understood, thus exploring other Bayesian and non-Bayesian methods would be a valuable contribution, especially in the elusive task of emotion recognition.

References

- Abdar, M., Pourpanah, F., Hussain, S., Reza-zadegan, D., Liu, L., Ghavamzadeh, M., ... Nahavandi, S. (2021, December). A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information Fusion*, 76, 243–297. Retrieved from <http://dx.doi.org/10.1016/j.inffus.2021.05.008> doi: 10.1016/j.inffus.2021.05.008
- Busso, C., Bulut, M., Lee, C.-C., Kazemzadeh, A., Mower, E., Kim, S., ... Narayanan, S. S. (2008, Nov). Iemocap: Interactive emotional dyadic motion capture database. *Language Resources and Evaluation*, 42(4), 335–359. doi: 10.1007/s10579-008-9076-6
- Chatterjee, A., Gupta, U., Chinnakotla, M. K., Srikanth, R., Galley, M., & Agrawal, P. (2019). Understanding emotions in text using deep learning and big data. *Computers in Human Behavior*, 93, 309–317. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0747563218306150> doi: <https://doi.org/10.1016/j.chb.2018.12.029>

- Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *CoRR*, *abs/1810.04805*. Retrieved from <http://arxiv.org/abs/1810.04805>
- Doğdu, C., Kessler, T., Schneider, D., Shadaydeh, M., & Schweinberger, S. R. (2022). A comparison of machine learning algorithms and feature sets for automatic vocal emotion recognition in speech. *Sensors*, *22*(19). Retrieved from <https://www.mdpi.com/1424-8220/22/19/7561> doi: 10.3390/s22197561
- Dzedzickis, A., Kaklauskas, A., & Bucinskas, V. (2020, Jan). Human emotion recognition: Review of sensors and methods. *Sensors*, *20*(3). doi: 10.3390/s20030592
- Dziugaite, G. K., Ben-David, S., & Roy, D. M. (2020). *Enforcing interpretability and its statistical impacts: Trade-offs between accuracy and interpretability*. Retrieved from <https://arxiv.org/abs/2010.13764>
- Ekman, P., & Friesen, W. V. (1971). Constants across cultures in the face and emotion. *Journal of Personality and Social Psychology*, *17*(2), 124–129. doi: 10.1037/h0030377
- Eyben, F., Weninger, F., Gross, F., & Schuller, B. (2013). Recent developments in opensmile, the munich open-source multimedia feature extractor. In *Proceedings of the 21st acm international conference on multimedia* (p. 835–838). New York, NY, USA: Association for Computing Machinery. Retrieved from <https://doi.org/10.1145/2502081.2502224> doi: 10.1145/2502081.2502224
- Eyben, F., Wöllmer, M., & Schuller, B. (2010). Opensmile: the munich versatile and fast open-source audio feature extractor. In *Proceedings of the 18th ACM international conference on Multimedia* (pp. 1459–1462).
- Gal, Y., & Ghahramani, Z. (2016). *Dropout as a bayesian approximation: Representing model uncertainty in deep learning*.
- Gandhi, B. S., Kumar, S., Victor, A., & Selvanambi, R. (2022). Transfer learning using bert comparative analysis of ml algorithms for opinion mining. In *2022 ieee 3rd global conference for advancement in technology (gcat)* (p. 1-12). doi: 10.1109/GCAT55367.2022.9971839
- Gawlikowski, J., Tassi, C. R. N., Ali, M., Lee, J., Humt, M., Feng, J., ... Zhu, X. X. (2022). *A survey of uncertainty in deep neural networks*. Retrieved from <https://arxiv.org/abs/2107.03342>
- Geethu, V., Vrindha, M. K., Anurenjan, P. R., Deepak, S., & Sreeni, K. G. (2023). Speech emotion recognition, datasets, features and models: A review. In *2023 international conference on control, communication and computing (iccc)* (p. 1-6). doi: 10.1109/ICCC57789.2023.10165567
- Ghosal, D., Majumder, N., Poria, S., Chhaya, N., & Gelbukh, A. (2019). *Dialoguecn: A graph convolutional neural network for emotion recognition in conversation*. Retrieved from <https://arxiv.org/abs/1908.11540>
- Gong, Y., Chung, Y.-A., & Glass, J. (2021). *Ast: Audio spectrogram transformer*. Retrieved from <https://arxiv.org/abs/2104.01778>
- Hu, D., Wei, L., & Huai, X. (2021). *Dialoguecrn: Contextual reasoning networks for emotion recognition in conversations*. Retrieved from <https://arxiv.org/abs/2106.01978>
- Kendall, A., & Gal, Y. (2017). What uncertainties do we need in bayesian deep learning for computer vision? *CoRR*, *abs/1703.04977*. Retrieved from <http://arxiv.org/abs/1703.04977>
- Li, P., Song, Y., McLoughlin, I., Guo, W., & Dai, L. (2018). An Attention Pooling Based Representation Learning Method for Speech Emotion Recognition. In *Proc. interspeech 2018* (pp. 3087–3091). doi: 10.21437/Interspeech.2018-1242
- Li, Y., Bell, P., & Lai, C. (2022). *Fusing asr outputs in joint training for speech emotion recognition*. Retrieved from <https://arxiv.org/abs/2110.15684>
- Li, Y., Wang, Y., Yang, X., & Im, S.-K. (2023, Oct). Speech emotion recognition based on graph-lstm

- neural network. *EURASIP Journal on Audio, Speech, and Music Processing*, 2023(1). doi: 10.1186/s13636-023-00303-9
- Majumder, N., Poria, S., Hazarika, D., Mihalcea, R., Gelbukh, A., & Cambria, E. (2019). *Dialoguenn: An attentive rnn for emotion detection in conversations*. Retrieved from <https://arxiv.org/abs/1811.00405>
- Meng, T., Shou, Y., Ai, W., Yin, N., & Li, K. (2023). *Deep imbalanced learning for multimodal emotion recognition in conversations*. Retrieved from <https://arxiv.org/abs/2312.06337>
- O'Shea, K., & Nash, R. (2015). *An introduction to convolutional neural networks*. Retrieved from <https://arxiv.org/abs/1511.08458>
- Pawłowski, M., Wróblewska, A., & Sysko-Romańczuk, S. (2023, Feb). Effective techniques for multimodal data fusion: A comparative analysis. *Sensors*, 23(5). doi: 10.3390/s23052381
- Pepino, L., Riera, P., Ferrer, L., & Gravano, A. (2020, May). Fusion approaches for emotion recognition from speech using acoustic and text-based features. In *Icassp 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. Retrieved from <http://dx.doi.org/10.1109/ICASSP40776.2020.9054709> doi: 10.1109/icassp40776.2020.9054709
- Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E., & Mihalcea, R. (2019). *Meld: A multimodal multi-party dataset for emotion recognition in conversations*. Retrieved from <https://arxiv.org/abs/1810.02508>
- Schuller, B., Steidl, S., Batliner, A., Vinciarelli, A., Scherer, K., Ringeval, F., ... Kim, S. (2013, 08). The interspeech 2013 computational paralinguistics challenge: Social signals, conflict, emotion, autism. In (p. 148-152). doi: 10.21437/Interspeech.2013-56
- Singh Tomar, P., Mathur, K., & Suman, U. (2023). Unimodal approaches for emotion recognition: A systematic review. *Cognitive Systems Research*, 77, 94-109. Retrieved from <https://www.sciencedirect.com/science/article/pii/S1389041722000626> doi: 10.1016/j.cogsys.2022.10.012
- Strivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56), 1929-1958. Retrieved from <http://jmlr.org/papers/v15/srivastava14a.html>
- TGDK, S., Selvarai, V., Raj, U., Raiu, V. P., & Prakash, J. (2021). Emotion detection using bi-directional lstm with an effective text pre-processing method. In *2021 12th international conference on computing communication and networking technologies (icccnt)* (p. 1-4). doi: 10.1109/ICCCNT51525.2021.9579844
- Trigeorgis, G., Ringeval, F., Brueckner, R., Marchi, E., Nicolaou, M. A., Schuller, B., & Zafeiriou, S. (2016). Adieu features? end-to-end speech emotion recognition using a deep convolutional recurrent network. In *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (p. 5200-5204). doi: 10.1109/ICASSP.2016.7472669
- User Guide - MoviePy 1.0.2 documentation. (n.d.). Retrieved from <https://zulko.github.io/moviepy/index.html>
- Valdenegro-Toro, M., & Mori, D. S. (2022). A deeper look into aleatoric and epistemic uncertainty disentanglement. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)* (p. 1508-1516). doi: 10.1109/CVPRW56347.2022.00157
- Waelen, R. (2024, Apr). Philosophical lessons for emotion recognition technology. *Minds and Machines*, 34(1). doi: 10.1007/s11023-024-09671-3
- Wang, F., Ding, Z., Xia, R., Li, Z., & Yu, J. (2021). *Multimodal Emotion-Cause Pair Extraction in Conversations*. doi: 10.48550/arXiv.2110.08020
- Yoon, S., Byun, S., & Jung, K. (2018). *Multimodal speech emotion recognition using audio and text*. Retrieved from <https://arxiv.org/abs/1810.04635>

A Architectural details

Audio-Only: Input([(None, 988)]) / Dense(None, 256) / Dropout(0.1) / Dense(None, 128) / Dropout(0.1) / Dense(None, 32) / Dropout(0.1) / Dense(None, 7), Dense(None, 7) / SamplingSoftmax(N=100)

Learning rate = 1.6854122029150702e-05

Text-Only: Input([(None, 768)]) / Dense(None, 256) / Dropout(0.2) / Dense(None, 128) / Dropout(0.2) / Dense(None, 64) / Dropout(0.2) / Dense(None, 7), Dense(None, 7) / SamplingSoftmax(N=150)

Learning rate = 0.000687224844798822

Early Fusion: Input([(None, 768)]), Input([(None, 988)]) / Concatenate() / Dense(None, 512) / Dropout(0.1) / Dense(None, 256) / Dropout(0.1) / Dense(None, 128) / Dropout(0.1) / Dense(None, 7), Dense(None, 7) / SamplingSoftmax(N=50)

Learning rate = 4.454196493570256e-05

Intermediate Fusion: {Input([(None, 768)]) / Dense(None, 400)}, {Input([(None, 988)]) / Dense(None, 400)} / Concatenate() / Dense(None, 512), Dropout(0.2) / Dense(None, 128) / Dropout(0.2) / Dense(None, 128) / Dropout(0.2) / Dense(None, 7), Dense(None, 7) / SamplingSoftmax(N=50)

Learning rate = 4.664276805358194e-06

Late Fusion: {Input([(None, 768)]) / Dense(None, 256) / Dropout(0.2) / Dense(None, 128) / Dropout(0.2) / Dense(None, 128) / Dropout(0.2)}, {Input([(None, 988)]) / Dense(None, 256) / Dropout(0.2) / Dense(None, 128) / Dropout(0.2) / Dense(None, 128) / Dropout(0.2)} / Concatenate() / Dense(None, 32) / Dropout(0.2) / Dense(None, 7), Dense(None, 7) / SamplingSoftmax(N=100)

Learning rate = 1.8549641064416335e-05